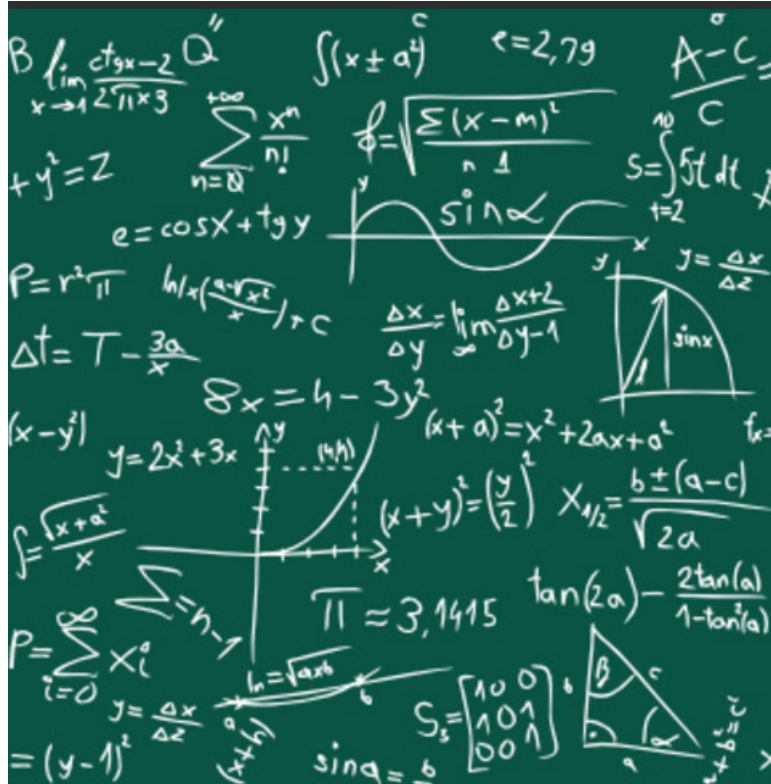


# Exercises: Curve sketching



- **Authors:** Fabio Furini and Gianluca Priori.
- **Book:** “Esercitazioni Analisi matematica 1”, Marco Bramanti, Esculapio.

### Exercise 1

For each of the following functions of the real variable  $x$ , determine:

1. the domain and the limits at its endpoints, pointing out any continuous extensions;
2. any local extrema and the intervals of monotonicity;
3. any inflection points and the intervals of convexity;
4. any asymptotes;
5. a qualitative graph that takes into account all the previous elements.

### Exercise 2

$$f(x) = x - \frac{1}{x}$$

The function  $f(x) = x - \frac{1}{x}$  is defined for  $x \neq 0$ , hence its natural domain  $D$  is given by

$$D = (-\infty, 0) \cup (0, +\infty).$$

We have

$$\lim_{x \rightarrow -\infty} f(x) = -\infty, \lim_{x \rightarrow 0^-} f(x) = +\infty, \lim_{x \rightarrow 0^+} f(x) = -\infty, \lim_{x \rightarrow +\infty} f(x) = +\infty,$$

in particular the line  $x = 0$  (the  $y$ -axis) is a vertical asymptote. From

$$\lim_{x \rightarrow \pm\infty} f(x) - x = \lim_{x \rightarrow \pm\infty} -\frac{1}{x} = 0$$

it then follows that the line  $y = x$  is an oblique asymptote as  $x \rightarrow \pm\infty$ . The presence of these asymptotes can also be obtained from analytic geometry: the curve with equation  $y = x - \frac{1}{x}$ , in implicit form  $x^2 - xy - 1 = 0$ , is a hyperbola whose asymptotes are precisely the lines  $x = 0$  and  $y = x$  (the asymptotes of a hyperbola are obtained by setting equal to 0 the homogeneous part of degree 2 of the equation).

The function  $f(x)$  is infinitely differentiable on its domain  $D$ . The first derivative is

$$f'(x) = 1 + \frac{1}{x^2}$$

and  $f'(x) > 0$  for every  $x \in D$ . Hence the function  $f(x)$  is strictly increasing on the interval  $(-\infty, 0)$  and also strictly increasing on the interval  $(0, +\infty)$ .

The second derivative is

$$f''(x) = -\frac{2}{x^3}$$

and it has the opposite sign of  $x$ . Hence the function  $f(x)$  is strictly convex on the interval  $(-\infty, 0)$  and strictly concave on the interval  $(0, +\infty)$ .

The position of the graph with respect to the oblique asymptote is easily deduced from  $f(x) - x = -1/x$ : we have  $f(x) > x$  for  $x < 0$  while  $f(x) < x$  for  $x > 0$ .

Now collect all the previous elements in a qualitative graph.

### Exercise 3

$$f(x) = \frac{\sqrt{x}}{\sqrt{x}-1}$$

The function  $f(x) = \frac{\sqrt{x}}{\sqrt{x}-1}$  is defined for  $x \geq 0$ ,  $x \neq 1$ , hence its natural domain  $D$  is given by

$$D = [0, 1) \cup (1, +\infty).$$

We have

$$\lim_{x \rightarrow 0} f(x) = f(0) = 0, \quad \lim_{x \rightarrow 1^-} f(x) = -\infty, \quad \lim_{x \rightarrow 1^+} f(x) = +\infty, \quad \lim_{x \rightarrow +\infty} f(x) = +1,$$

in particular the line  $x = 1$  is a vertical asymptote and the line  $y = 1$  is a horizontal asymptote as  $x \rightarrow +\infty$ .

The function  $f(x)$  is continuous on  $D$  and infinitely differentiable on  $D \setminus \{0\}$ . At the point  $x = 0$  the difference quotient has limit

$$\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} = \lim_{x \rightarrow 0} \frac{\sqrt{x}}{x(\sqrt{x} - 1)} = -\lim_{x \rightarrow 0} \frac{\sqrt{x}}{x} = -\lim_{x \rightarrow 0} \frac{1}{\sqrt{x}} = -\infty$$

hence the function  $f(x)$  is not differentiable at  $x = 0$  and at the point  $(0, 0)$  the graph has the  $y$ -axis as a vertical tangent.

For  $x \in D \setminus \{0\}$  the first derivative is

$$f'(x) = \frac{-1}{2\sqrt{x}(\sqrt{x}-1)^2}$$

and  $f'(x) < 0$  for every  $x$ . Hence the function  $f(x)$  is strictly decreasing on the interval  $[0, 1)$  and also strictly decreasing on the interval  $(1, +\infty)$ .

The second derivative is

$$f''(x) = \frac{1}{4} \frac{3\sqrt{x} - 1}{x\sqrt{x}(\sqrt{x}-1)^3}$$

which vanishes at  $x = 1/9$ , is positive for  $0 < x < 1/9$ , negative for  $1/9 < x < 1$ , and positive again for  $x > 1$ . Hence the function  $f(x)$  is strictly convex on the interval  $(0, 1/9)$ , strictly concave on the interval  $(1/9, 1)$ , and strictly convex again on the interval  $(1, +\infty)$ . At  $x = 1/9$  there is an inflection point: at the point  $(1/9, -1/2)$  the graph crosses from above to below the tangent line.

Now collect all the previous elements in a qualitative graph.

#### Exercise 4

$$f(x) = \sqrt{x^2 - 1} - \sqrt{x^2 + 1}$$

The function  $f(x) = \sqrt{x^2 - 1} - \sqrt{x^2 + 1}$  is defined for  $x \leq -1$ ,  $x \geq 1$ , hence its natural domain  $D$  is given by

$$D = (-\infty, -1] \cup [1, +\infty).$$

The function under consideration is even: the following study could be restricted to the interval  $[1, +\infty)$ . We have

$$\lim_{x \rightarrow \pm\infty} f(x) = \lim_{x \rightarrow \pm\infty} \frac{-2}{\sqrt{x^2 - 1} + \sqrt{x^2 + 1}} = 0$$

in particular the line  $y = 0$  (the  $x$ -axis) is a horizontal asymptote as  $x \rightarrow \pm\infty$ . We then have

$$\lim_{x \rightarrow -1} f(x) = f(-1) = -\sqrt{2}, \quad \lim_{x \rightarrow 1} f(x) = f(1) = -\sqrt{2}.$$

The function  $f(x)$  is continuous on  $D$  and infinitely differentiable on  $D \setminus \{-1, 1\}$ . For  $x \in D \setminus \{-1, 1\}$  the first derivative is

$$f'(x) = \frac{x}{\sqrt{x^2 - 1}} - \frac{x}{\sqrt{x^2 + 1}} = \frac{x(\sqrt{x^2 + 1} - \sqrt{x^2 - 1})}{\sqrt{x^4 - 1}}$$

and it has the same sign as  $x$ . Hence the function  $f(x)$  is strictly decreasing on the interval  $(-\infty, -1)$  and strictly increasing on the interval  $(1, +\infty)$ .

At the points  $x = \pm 1$  we have

$$\lim_{x \rightarrow \pm 1} f'(x) = \pm\infty$$

hence the function  $f(x)$  is not differentiable at  $x = \pm 1$  and at the points  $(\pm 1, -\sqrt{2})$  the graph has a vertical tangent.

The second derivative is

$$f''(x) = -\frac{1}{(x^2 - 1)\sqrt{x^2 - 1}} - \frac{1}{(x^2 + 1)\sqrt{x^2 + 1}}$$

and it is negative for every  $x$ . Hence the function  $f(x)$  is strictly concave on the interval  $(-\infty, -1)$  and on the interval  $(1, +\infty)$ .

Now collect all the previous elements in a qualitative graph, respecting in particular the symmetry of the graph with respect to the  $y$ -axis (even symmetry).

#### Exercise 5

$$f(x) = \frac{\sqrt{2x-1}}{\log(2x-1)}$$

The function  $f(x) = \frac{\sqrt{2x-1}}{\log(2x-1)}$  is defined for  $2x-1 > 0$ ,  $2x-1 \neq 1$ , hence its natural domain  $D$  is given by

$$D = (1/2, 1) \cup (1, +\infty).$$

We have

$$\lim_{x \rightarrow 1/2} f(x) = 0, \quad \lim_{x \rightarrow 1^\pm} f(x) = \pm\infty, \quad \lim_{x \rightarrow +\infty} f(x) = \lim_{y \rightarrow +\infty} \frac{y^{1/2}}{\log y} = +\infty$$

in particular the line  $x = 1$  is a vertical asymptote. The order of infinity of  $f(x)$  as  $x \rightarrow +\infty$  is lower than that of  $\sqrt{x}$ , which rules out an oblique asymptote (the order of infinity of functions with an oblique asymptote is that of  $x$ ).

The function  $f(x)$  is continuous on  $D$ ; the limit  $\lim_{x \rightarrow 1/2} f(x) = 0$  allows us to extend the function continuously also to  $x = 1/2$  by setting  $f(1/2) = 0$ .

The function is infinitely differentiable on  $D$ . The first derivative is

$$f'(x) = \frac{\log(2x-1) - 2}{\sqrt{2x-1} \log^2(2x-1)}.$$

At the point  $x = 1/2$ , the difference quotient of the continuous extension has limit

$$\begin{aligned} \lim_{x \rightarrow 1/2} \frac{f(x) - f(1/2)}{x - 1/2} &= \lim_{x \rightarrow 1/2} \frac{\sqrt{2x-1}}{(x - 1/2) \log(2x-1)} = \\ &= \sqrt{2} \lim_{x \rightarrow 1/2} \frac{1}{\sqrt{x-1/2} \log(2x-1)} = \sqrt{2} \lim_{y \rightarrow 0} \frac{1}{y^{1/2} \log y} = -\infty. \end{aligned}$$

The continuous extension of  $f$  is not differentiable at  $x = 1/2$ . At the point  $(1/2, 0)$  its graph has a vertical tangent.

Going back to  $f'(x)$  at the points of  $D$ , we have  $f'(x) > 0$  for  $x > (e^2 + 1)/2$ ,  $f'(x) < 0$  for  $x < (e^2 + 1)/2$ ,  $f'(x) = 0$  for  $x = (e^2 + 1)/2$ . Hence the function  $f(x)$  is strictly decreasing on the interval  $(1/2, 1)$  and on the interval  $(1, (e^2 + 1)/2)$ , and strictly increasing on the interval  $((e^2 + 1)/2, +\infty)$ . The point  $x = (e^2 + 1)/2$  is a local minimum point with value  $f((e^2 + 1)/2) = e/2$ .

The second derivative is

$$f''(x) = -\frac{\log^2(2x-1) - 8}{(2x-1)\sqrt{2x-1} \log^3(2x-1)}$$

and it is positive for  $x \in (1/2, (e^{-\sqrt{8}} + 1)/2)$ , negative for  $x \in ((e^{-\sqrt{8}} + 1)/2, 1)$ , positive for  $x \in (1, (e^{\sqrt{8}} + 1)/2)$ , negative for  $x \in ((e^{\sqrt{8}} + 1)/2, +\infty)$ . It follows that  $f$  is strictly convex for  $x \in (1/2, (e^{-\sqrt{8}} + 1)/2)$ , strictly concave for  $x \in ((e^{-\sqrt{8}} + 1)/2, 1)$ , strictly convex for  $x \in (1, (e^{\sqrt{8}} + 1)/2)$ , strictly concave for  $x \in ((e^{\sqrt{8}} + 1)/2, +\infty)$ . The points  $x = (e^{\pm\sqrt{8}} + 1)/2$  are inflection points with respective values  $f((e^{\pm\sqrt{8}} + 1)/2) = \pm e^{\pm\sqrt{2}}/\sqrt{8}$ .

Now collect all the previous elements in a qualitative graph.

### Exercise 6

$$f(x) = xe^{\frac{1}{x}}$$

The function  $f(x) = xe^{\frac{1}{x}}$  is defined for  $x \neq 0$ , hence its natural domain  $D$  is given by

$$D = (-\infty, 0) \cup (0, +\infty).$$

We have

$$\lim_{x \rightarrow -\infty} f(x) = -\infty, \quad \lim_{x \rightarrow 0^-} f(x) = 0,$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{y \rightarrow +\infty} \frac{e^y}{y} = +\infty, \quad \lim_{x \rightarrow +\infty} f(x) = +\infty$$

in particular the line  $x = 0$  (the  $y$ -axis) is a vertical asymptote. As  $x \rightarrow \pm\infty$ , from  $e^y = 1 + y + o(y)$  as  $y \rightarrow 0$ , we obtain

$$f(x) = xe^{\frac{1}{x}} = x \left( 1 + \frac{1}{x} + o\left(\frac{1}{x}\right) \right) = x + 1 + o(1)$$

hence the line  $y = x + 1$  is an oblique asymptote as  $x \rightarrow \pm\infty$ .

The function  $f(x)$  is continuous on  $D$ ; the limit  $\lim_{x \rightarrow 0^-} f(x) = 0$  allows us to extend the function continuously from the left to  $x = 0$  by setting  $f(0) = 0$ .

The function is infinitely differentiable on  $D$ . The first derivative is

$$f'(x) = \left( 1 - \frac{1}{x} \right) e^{\frac{1}{x}} = \frac{x-1}{x} e^{\frac{1}{x}}.$$

At the point  $x = 0$ , the left difference quotient of the extension has limit

$$\lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x} = \lim_{x \rightarrow 0^-} e^{\frac{1}{x}} = 0$$

The left continuous extension of  $f$  is left-differentiable at  $x = 0$  with  $f'_-(0) = 0$ .

The negative  $x$  half-axis is a tangent half-line to the graph at the point  $(0, 0)$ .

Going back to  $f'(x)$  at the points of  $D$ , we have  $f'(x) > 0$  for  $x < 0$ ,  $f'(x) < 0$  for  $0 < x < 1$ , again  $f'(x) > 0$  for  $x > 1$ , and  $f'(x) = 0$  for  $x = 1$ . Hence the function  $f(x)$  is strictly increasing on the interval  $(-\infty, 0)$ , strictly decreasing on the interval  $(0, 1)$ , and strictly increasing again on the interval  $(1, +\infty)$ . The point  $x = 1$  is a local minimum point with value  $f(1) = e$ .

The second derivative is

$$f''(x) = \frac{1}{x^3} e^{\frac{1}{x}}$$

and it has the same sign as  $x$ . It follows that  $f$  is strictly convex for  $x \in (0, +\infty)$  and strictly concave for  $x \in (-\infty, 0)$ .

Now collect all the previous elements in a qualitative graph.

### Exercise 7

$$f(x) = e^{\frac{1-|x|}{1+x}}$$

The function  $f(x) = e^{\frac{1-|x|}{1+x}}$  is defined for  $x \neq -1$ , hence its natural domain  $D$  is given by

$$D = (-\infty, -1) \cup (-1, +\infty).$$

On the set  $(-\infty, -1) \cup (-1, 0]$  the function is constant:  $f(x) = e^{\frac{1+x}{1+x}} = e$  for every  $x \leq 0$ ,  $x \neq -1$ . Obviously we can extend the function continuously to  $x = -1$  by defining  $f(-1) = e$ . We then have

$$\lim_{x \rightarrow +\infty} f(x) = \lim_{x \rightarrow +\infty} e^{\frac{1-x}{1+x}} = e^{-1} = \frac{1}{e}$$

in particular the line  $y = 1/e$  is a horizontal asymptote as  $x \rightarrow +\infty$ .

The function  $f(x)$  is continuous on  $D$ .

The function is infinitely differentiable on  $D \setminus \{0\}$ . The first derivative is obviously 0 for  $x < 0$ , while it is

$$f'(x) = -\frac{2}{(x+1)^2} e^{\frac{1-x}{1+x}} \text{ for } x > 0.$$

At the point  $x = 0$ , the left derivative is 0 while

$$\lim_{x \rightarrow 0^+} f'(x) = -2e.$$

The function is not differentiable at  $x = 0$  since  $f'_-(0) \neq f'_+(0)$ . The graph has a corner point at  $(0, e)$ : on the left it consists of the half-line  $y = e$ ,  $x \leq 0$ , while the right branch has tangent half-line  $y = -2ex + e$ ,  $x \geq 0$ .

We then have  $f'(x) < 0$  for every  $x > 0$ , hence  $f$  is strictly decreasing on the interval  $(0, +\infty)$ .

Since the function is constant for  $x \leq 0$ , we only need to compute the second derivative for  $x > 0$ , where it is

$$f''(x) = \frac{4x+8}{(x+1)^4} e^{\frac{1-x}{1+x}}$$

and it is positive. It follows that  $f$  is strictly convex for  $x \in (0, +\infty)$ .

Now collect all the previous elements in a qualitative graph.

### Exercise 8

$$f(x) = \left| \frac{\log x}{x} \right|$$

The function  $f(x) = \left| \frac{\log x}{x} \right|$  is defined for  $x > 0$ , hence its natural domain  $D$  is given by

$$D = (0, +\infty)$$

where its expression can be simplified to

$$f(x) = \frac{|\log x|}{x}, \quad x \in D.$$

Keep in mind, now and in what follows, the sign of  $\log x$ , which gives  $|\log x| = -\log x$  for  $0 < x < 1$  and  $|\log x| = \log x$  for  $x > 1$ . We have

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{-\log x}{x} = +\infty, \quad \lim_{x \rightarrow +\infty} f(x) = \lim_{x \rightarrow +\infty} \frac{\log x}{x} = 0$$

in particular the line  $x = 0$  (the  $y$ -axis) is a vertical asymptote while the line  $y = 0$  (the  $x$ -axis) is a horizontal asymptote as  $x \rightarrow +\infty$ .

The function  $f(x)$  is continuous on  $D$ , it clearly takes only non-negative values, and the point  $x = 1$ , where  $f(1) = 0$ , is an absolute minimum point.

The function is infinitely differentiable on  $D \setminus \{1\}$ . The first derivative is

$$f'(x) = -\frac{1 - \log x}{x^2} \text{ for } 0 < x < 1; \quad f'(x) = \frac{1 - \log x}{x^2} \text{ for } x > 1.$$

At the point  $x = 1$ , the left derivative is

$$\lim_{x \rightarrow 1^-} f'(x) = \lim_{x \rightarrow 1^-} -\frac{1 - \log x}{x^2} = -1,$$

while the right derivative is

$$\lim_{x \rightarrow 1^+} f'(x) = \lim_{x \rightarrow 1^+} \frac{1 - \log x}{x^2} = 1$$

The function is not differentiable at  $x = 1$  since  $f'_-(1) \neq f'_+(1)$ . The graph has a corner point at  $(1, 0)$ : on the left the tangent half-line has equation  $y = -x + 1$ , on the right  $y = x - 1$ .

We then have  $f'(x) < 0$  for every  $x \in (0, 1)$ , hence  $f$  is strictly decreasing on the interval  $(0, 1)$ . We have  $f'(x) > 0$  for every  $x \in (1, e)$  and  $f'(x) < 0$  for every  $x \in (e, +\infty)$ ,  $f'(e) = 0$ , hence  $f$  is strictly increasing on the interval  $(1, e)$  and strictly decreasing on the interval  $(e, +\infty)$ ; the point  $x = e$  is a relative maximum point with value  $f(e) = 1/e$ .

On  $D \setminus \{1\}$  the second derivative is

$$f''(x) = \frac{3 - 2\log x}{x^3} \text{ for } 0 < x < 1; \quad f''(x) = \frac{2\log x - 3}{x^3} \text{ for } x > 1.$$

It follows that  $f$  is strictly convex for  $x \in (0, 1)$  and on  $(e^{3/2}, +\infty)$ ; strictly concave on  $(1, e^{3/2})$ . The point  $x = e^{3/2}$  is an inflection point with value  $f(e^{3/2}) = 3/2e^{3/2}$ . Now collect all the previous elements in a qualitative graph.

#### Exercise 9

$$f(x) = \sqrt{1 - |e^{2x} - 1|}$$

The function  $f(x) = \sqrt{1 - |e^{2x} - 1|}$  is defined for  $|e^{2x} - 1| \leq 1$ , hence for  $-1 \leq e^{2x} - 1 \leq 1$ , from which  $0 \leq e^{2x} \leq 2$  and finally  $x \leq (\log 2)/2$ , since  $e^{2x} > 0$  for every  $x$ . The natural domain  $D$  is given by

$$D = (-\infty, (\log 2)/2].$$

Keep in mind, now and in what follows, the sign of  $e^{2x} - 1$ , which gives  $|e^{2x} - 1| = 1 - e^{2x}$  for  $x < 0$  and  $|e^{2x} - 1| = e^{2x} - 1$  for  $x > 0$ . In particular

$$f(x) = \sqrt{e^{2x}} = e^x \text{ for } x < 0; \quad f(x) = \sqrt{2 - e^{2x}} \text{ for } 0 \leq x \leq (\log 2)/2.$$

The behavior of  $e^x$  for  $x < 0$  is well known; in particular it follows that

$$\lim_{x \rightarrow -\infty} f(x) = 0$$

(the  $x$ -axis is a horizontal asymptote as  $x \rightarrow -\infty$ ) and that  $f$  is strictly increasing on  $(-\infty, 0)$ . We then have

$$\lim_{x \rightarrow (\log 2)/2} f(x) = f((\log 2)/2) = 0.$$

The function  $f(x)$  is continuous on  $D$ , it clearly takes only non-negative values, and the point  $x = (\log 2)/2$ , where  $f((\log 2)/2) = 0$ , is an absolute minimum point. The function is infinitely differentiable on  $D \setminus \{0, (\log 2)/2\}$ . The first derivative is

$$f'(x) = e^x \text{ for } x < 0; \quad f'(x) = -\frac{e^{2x}}{\sqrt{2 - e^{2x}}} \text{ for } 0 < x < (\log 2)/2.$$

At the point  $x = 0$ , the left derivative is

$$\lim_{x \rightarrow 0^-} f'(x) = \lim_{x \rightarrow 0^-} e^x = 1,$$

while the right derivative is

$$\lim_{x \rightarrow 0^+} f'(x) = \lim_{x \rightarrow 0^+} -\frac{e^{2x}}{\sqrt{2 - e^{2x}}} = -1$$

The function is not differentiable at  $x = 0$  since  $f'_-(0) \neq f'_+(0)$ . The graph has a corner point at  $(0, 1)$ : on the left the tangent half-line has equation  $y = x + 1$ , on the right  $y = -x + 1$ .

At the point  $x = (\log 2)/2$  we have

$$\lim_{x \rightarrow (\log 2)/2} f'(x) = \lim_{x \rightarrow (\log 2)/2} -\frac{e^{2x}}{\sqrt{2 - e^{2x}}} = -\infty.$$

The function is not differentiable at  $x = (\log 2)/2$ . At the point  $((\log 2)/2, 0)$  the graph has a vertical tangent.

We then have  $f'(x) < 0$  for every  $x \in (0, (\log 2)/2)$ , hence  $f$  is strictly decreasing on the interval  $(0, (\log 2)/2)$ . The point  $x = 0$  is an (absolute) maximum point with value  $f(0) = 1$ .

On  $(-\infty, 0)$  the function coincides with  $e^x$ , and it is well known that the exponential function is strictly convex. We then have

$$f''(x) = -\frac{e^{2x}(4 - e^{2x})}{(2 - e^{2x})^{3/2}} \text{ for } 0 < x < (\log 2)/2$$

which is negative on that interval. It follows that  $f$  is strictly concave for  $x \in (0, (\log 2)/2)$ .

Now collect all the previous elements in a qualitative graph.

(i) The function  $f(x) = x - \arctan x$  has natural domain  $D = \mathbb{R}$  with

$$\lim_{x \rightarrow \pm\infty} f(x) = \pm\infty.$$

It is an odd function, which would allow us to study it only for  $x \geq 0$ .

From

$$\lim_{x \rightarrow \pm\infty} f(x) - x = \lim_{x \rightarrow \pm\infty} -\arctan x = \mp \frac{\pi}{2}$$

it then follows that the function has the oblique asymptotes

$$y = x + \pi/2, x \rightarrow -\infty; \quad y = x - \pi/2, x \rightarrow +\infty.$$

The function is infinitely differentiable on all of  $\mathbb{R}$ . The first derivative is

$$f'(x) = 1 - \frac{1}{x^2 + 1} = \frac{x^2}{x^2 + 1}$$

with  $f'(0) = 0$  and  $f'(x) > 0$  for every  $x \neq 0$ . It follows that  $f$  is strictly increasing on all of  $\mathbb{R}$  and that the point  $x = 0$  is an inflection point with horizontal tangent, with corresponding value  $f(0) = 0$ .

The second derivative is

$$f''(x) = \frac{2x}{(x^2 + 1)^2}$$

and it has the same sign as  $x$ . In particular  $f$  is strictly concave on  $(-\infty, 0)$  and strictly convex on  $(0, +\infty)$ .

Now collect all the previous elements in a qualitative graph, respecting the central symmetry of the graph with respect to the origin (odd symmetry).

### Exercise 10

$$f(x) = x - \arctan x$$

The function  $f(x) = x - \arctan x$  has natural domain  $D = \mathbb{R}$  with

$$\lim_{x \rightarrow \pm\infty} f(x) = \pm\infty.$$

It is an odd function, which would allow us to study it only for  $x \geq 0$ .

From

$$\lim_{x \rightarrow \pm\infty} f(x) - x = \lim_{x \rightarrow \pm\infty} -\arctan x = \mp \frac{\pi}{2}$$

it then follows that the function has the oblique asymptotes

$$y = x + \pi/2, x \rightarrow -\infty; \quad y = x - \pi/2, x \rightarrow +\infty.$$

The function is infinitely differentiable on all of  $\mathbb{R}$ . The first derivative is

$$f'(x) = 1 - \frac{1}{x^2 + 1} = \frac{x^2}{x^2 + 1}$$

with  $f'(0) = 0$  and  $f'(x) > 0$  for every  $x \neq 0$ . It follows that  $f$  is strictly increasing on all of  $\mathbb{R}$  and that the point  $x = 0$  is an inflection point with horizontal tangent, with corresponding value  $f(0) = 0$ .

The second derivative is

$$f''(x) = \frac{2x}{(x^2 + 1)^2}$$

and it has the same sign as  $x$ . In particular  $f$  is strictly concave on  $(-\infty, 0)$  and strictly convex on  $(0, +\infty)$ .

Now collect all the previous elements in a qualitative graph, respecting the central symmetry of the graph with respect to the origin (odd symmetry).

### Exercise 11

$$f(x) = 1 - x^2 + \log|x|$$

The function  $f(x) = 1 - x^2 + \log|x|$  is defined for  $x \neq 0$ , hence its natural domain  $D$  is given by

$$D = (-\infty, 0) \cup (0, +\infty).$$

It is an even function, which would allow us to study it only for  $x > 0$ .

From  $f(x) \sim -x^2$  as  $x \rightarrow \pm\infty$  we have

$$\lim_{x \rightarrow \pm\infty} f(x) = -\infty$$

and we deduce that the function has no oblique asymptotes. We then have

$$\lim_{x \rightarrow 0^\pm} f(x) = -\infty,$$

in particular the line  $x = 0$  (the  $y$ -axis) is a vertical asymptote.

The function is infinitely differentiable on all of  $D$ . The first derivative is

$$f'(x) = -2x + \frac{1}{x} = \frac{1 - 2x^2}{x}.$$

Analyzing its sign for  $x > 0$ , we have  $f'(\sqrt{1/2}) = 0$ ,  $f'(x) > 0$  for  $0 < x < \sqrt{1/2}$ ,  $f'(x) < 0$  for  $x > \sqrt{1/2}$ . It follows that  $f$  is strictly increasing on  $(0, \sqrt{1/2})$ , strictly decreasing on  $(\sqrt{1/2}, +\infty)$ , and that the point  $x = \sqrt{1/2}$  is an (absolute) maximum point with corresponding value  $f(\sqrt{1/2}) = 1/2 - (\log 2)/2$ . In particular the graph meets the  $x$ -axis at two points with positive abscissa. One of these points is  $x = 1$ ; the other has abscissa in the interval  $(0, \sqrt{1/2})$  and can be approximated, if needed, with one of the algorithms studied for finding the zeros of smooth functions.

The study of monotonicity and extrema of  $f$  for  $x < 0$  follows by even symmetry: the point  $x = -\sqrt{1/2}$  is an (absolute) maximum point with corresponding value  $f(-\sqrt{1/2}) = 1/2 - (\log 2)/2$ , etc., etc.

The second derivative is

$$f''(x) = -2 - \frac{1}{x^2}$$

and it is negative for every  $x \in D$ . In particular  $f$  is strictly concave both on  $(-\infty, 0)$  and on  $(0, +\infty)$ .

Now collect all the previous elements in a qualitative graph, respecting the axial symmetry of the graph with respect to the  $y$ -axis (even symmetry).